

DG Interns Hub – Cybersecurity Internship



Week 1 Assignment Report: Networking Fundamentals

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TASK 1: OSI MODEL

- OSI MODEL (OPEN SYSTEMS INTERCONNECTION) HAS 7 LAYERS is a conceptual framework developed by the International Organization for Standardization (ISO) to explain how communication takes place between different devices over a network.
- The OSI model divides the network into 7 different layers. These are made for better network understanding, troubleshooting etc.
- Each layer has a specific function to do.
- The 7 different layers of OSI models are:

Layer 7: Application Layer

- The Application Layer is the seventh and highest layer of the OSI model. It provides network services directly to applications that users interact with.
- It allows software applications to communicate over a network.
- Ex: HTTP, HTTPS, FTP

Layer 6: Presentation Layer

- The Presentation Layer is responsible for making sure that data sent by one system can be correctly understood by another system.

- The main functions of it are data encryption and decryption, data compression
- Ex: Turns readable data into encrypted version and transmits the data.

Layer 5: Session Layer

- The Session Layer establishes, manages, and terminates communication sessions between applications.
- A session is a period of communication between two systems.
- The main functions are to establish a session, maintain the session, synchronize communication, and terminate the session.
- Example: During a video conference, the session layer helps establish, maintain, and terminate the communication session.

Layer 4: Transport Layer

- The Transport Layer provides end-to-end communication between devices.
- The main functions are segmentation, reliable data delivery, error detection, flow control, and data recovery.

flow control and managing communication between applications

- It has protocols those are TCP (Transmission Control Panel) and UDP (User Datagram Protocol)
- Ex: When downloading a PDF from a website, TCP helps ensure that all the pieces of the file arrive correctly.

Layer 3: Network Layer

- The Network Layer is responsible for delivering packets between different networks.
- Its major functions include Logical addressing, routing, packet forwarding, path selection.
- The most important protocol associated with this layer is Internet Protocol.
- Ex: sending the message from one country to another. The message is checked in various servers.

Layer 2 : Data Link Layer

- The Data Link Layer provides reliable communication between devices that are directly connected within the same network segment.

- Its main functions are framing, MAC addressing, error detection, and controlling access to the transmission medium.
- Example: A computer sends an Ethernet frame to a switch, which examines the frame and forwards it to the appropriate device.

Layer 1: Physical Layer

- The Physical Layer is the lowest layer of the OSI model. It deals with the actual transmission of raw bits 0s and 1s through a physical or wireless medium.
- Its characteristics are cables, connectors and electric signals.
- Ex: Ethernet cables, Fiber-optic cables, Network connectors

TASK 2: TCP/IP

- The TCP/IP (Transmission Control Protocol/Internet Protocol) Model is a networking framework used to describe how devices communicate over the Internet and other computer networks.

- the TCP/IP model commonly has 4 layers.
They are

Layer 4: Application Layer

- The Application Layer is the highest layer of the TCP/IP model. It provides network services to applications that users interact with.
- It combines the functions of the Application, Presentation, and Session layers of the OSI model.
- Its main functions are Providing network services to applications, Data representation and formatting, Encryption-related functions, Supporting different network protocols
- Its common protocols are HTTP, HTTPS, FTP
- Ex: when we type browser name on the search engine, the application layers use protocols like DNS and HTTPS to connect to google servers.

Layer 3: Transport Layer

- The Transport Layer provides communication between applications running on different devices.
- It corresponds to the Transport Layer of the OSI model.

- Its main functions are end to end communication, Data segmentation, reliable delivery, error control and flow control.
- Its main protocols are TCP and UDP
- Ex: When you download a document from a website, TCP helps ensure that the complete document reaches your computer correctly.

Layer 2: Internet Layer

- The Internet Layer is responsible for delivering packets between different networks. It corresponds approximately to the Network Layer of the OSI model.
- Its main functions are logical addressing, packet routing, packet forwarding
- The important protocols are IP, ICMP, ARP
- Ex: When you send data from your computer to a website server, routers use destination IP addresses to determine where packets should be forwarded.

Layer 1: Network Access Layer

- The Network Access Layer is the lowest layer of the TCP/IP model.

- Its main functions are Physical transmission of data, Frames, MAC addresses and Network hardware
- Ex: Ethernet, Wi-Fi etc

Difference between OSI model and TCP/IP model

OSI MODEL

TCP/IP MODEL

Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Has 7 clearly defined layers	Has 4 layers with some functions combined
Developed by ISO	Developed through DARPA/DoD research community
Mainly used as a reference model for learning, design, and troubleshooting	Widely used in real-world Internet networking
Protocol-independent	Built around networking protocols

TASK 3: NETWORK PROTOCOLS

- A network protocol is a set of rules that defines how devices communicate and exchange data over a network.
- HTTP / HTTPS: Governs communication between web browsers and web servers
- Ex: When you visit a website such as a news website, your browser requests web pages, images, and other resources from the server.
- FTP: File Transfer Protocol is a protocol designed for transferring files between computers over a network.
- Ex: Suppose a web developer needs to upload website files to a hosting server. FTP can be used to transfer those files.
- DNS: Domain Name System is a system that translates human-readable domain names into IP addresses.
- Ex: when a website name is typed the device needs to determine the IP address associated with that domain.

- DHCP: Dynamic Host Configuration Protocol automatically provides network configuration information to devices joining a network.
- Ex: When you connect your smartphone or laptop to your home Wi-Fi, you normally don't manually enter an IP address.

TASK 4: IP ADDRESSING

- IP Addressing is a logical address assigned to a device connected to a network. It is used to identify the device and help deliver data to the correct destination.
- IP Versions
 - IPv4: IPv4 is the most commonly encountered IP addressing format. An IPv4 address consists of 32 bits, divided into four groups called octets.
 - IPV6: IPv6 was developed to provide a much larger number of addresses than IPv4. An IPv6 address consists of 128 bits.

Public vs Private IP Address

- A public IP address is an IP address that is used to identify a network or device on the public Internet.
- Public IP addresses need to be globally unique within the Internet's addressing system.
- A private IP address is used inside a private network, such as a home, school, office, or organization.
- Ex: ISP-assigned address for public IP address and 192.168.1.10 is for private address

Basic subnetting concept

- Subnetting is the process of dividing a larger IP network into smaller logical networks called subnets.
- Subnetting helps organizations use IP addresses efficiently and divide networks into manageable sections.
- A subnet mask is used with an IPv4 address to determine which portion represents the network and which portion represents the host.
- Example: For 192.168.1.0/24, the network address is 192.168.1.0, usable host addresses are 192.168.1.1–192.168.1.254, and the broadcast address is 192.168.1.255.

TASK 5: NETWORK COMMANDS

- Ping google.com

```
Microsoft Windows [Version 10.0.26200.9160]
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C:\Users\volgarping google.com

Pinging google.com [142.250.66.14] with 32 bytes of data:
Reply from 142.250.66.14: bytes=32 time=17ms TTL=118
Reply from 142.250.66.14: bytes=32 time=19ms TTL=118
Reply from 142.250.66.14: bytes=32 time=18ms TTL=118
Reply from 142.250.66.14: bytes=32 time=16ms TTL=118

Ping statistics for 142.250.66.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 19ms, Average = 17ms

C:\Users\volgar>
```

- IP Configuration

```
Windows PowerShell
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PS C:\Users\volgar> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix . : 
    Link-Local IPv6 Address . . . . . : fe80::71b4:a38e:81af:e980%18
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : domain.name
    Link-Local IPv6 Address . . . . . : fe80::6a73:4c8:a5f0:ad85%7
    IPv4 Address. . . . . : 192.168.2.7
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::217:7c7f:f64a:85ad%7
    192.168.2.1

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 
PS C:\Users\volgar>
```

■ Tracert google.com

```

Windows PowerShell
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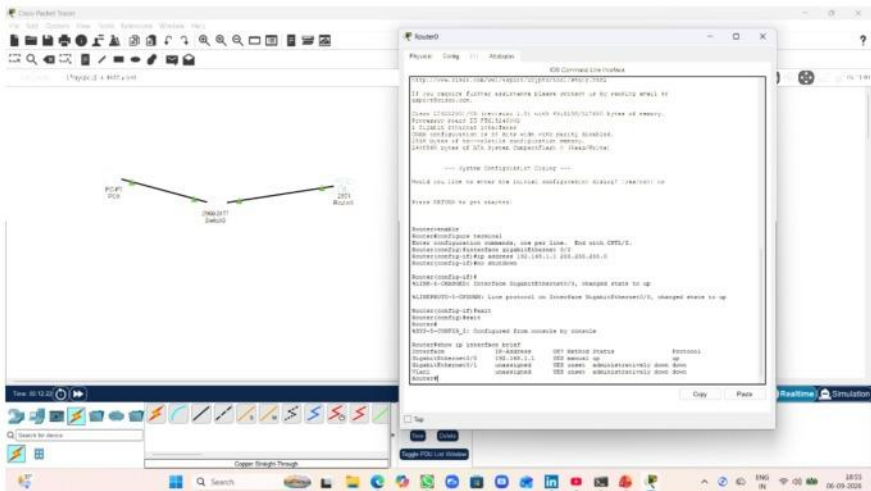
PS C:\Users\bolgur> tracert google.com

Tracing route to google.com [142.250.77.142]
over a maximum of 30 hops:
  0  2 ms  2 ms  2 ms  192.168.2.1
  1  4 ms  4 ms  4 ms  10.130.96.1
  2  6 ms  5 ms  5 ms  breadband.actcorp.in [183.83.249.101]
  3  *      *      *      Request timed out.
  4  *      *      *      Request timed out.
  5  *      *      *      Request timed out.
  6  28 ms  16 ms  18 ms  breadband.actcorp.in [183.82.14.78]
  7  16 ms  16 ms  16 ms  72.14.243.242
  8  21 ms  19 ms  16 ms  216.239.51.91
  9  17 ms  16 ms  16 ms  142.251.55.61
 10  26 ms  16 ms  19 ms  maa85a16-in-f36.1e108.net [142.250.77.142]

Trace complete.
PS C:\Users\bolgur>

```

TASK 6: NETWORK SIMULATIONS



■ Ping 192.168.1.1

